Section 5.3 - Limits at Infinity

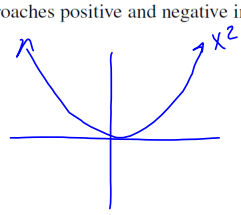
Example 1: Let's examine the behavior of x^2 and $\frac{1}{x^2}$ as x approaches positive and negative infinity.

$\lim_{x\to\infty} x^2 = \infty$

$\lim_{x\to-\infty} x^2 = \infty$

$\lim_{x\to\infty} \frac{1}{x^2} = 0$

$\lim_{x\to-\infty} \frac{1}{x^2} = 0$



Thus, $y=0$ is a horizontal asymptote of $f(x)=\frac{1}{x^2}$

Limits of Power Functions at Infinity

If p is a positive real number and k is any real constant, then

1. $\lim_{x\to\infty} kx^p = \pm\infty$
2. $\lim_{x\to-\infty} kx^p = \pm\infty$
3. $\lim_{x\to\infty} \frac{k}{x^p} = 0$
4. $\lim_{x\to-\infty} \frac{k}{x^p} = 0$

Definition: A line $y=b$ is a **horizontal asymptote** of the graph of $y=f(x)$ if $f(x)$ approaches b as either x increases or decreases without bound.

$\lim_{x\to-\infty} f(x) = b$ AND/OR $\lim_{x\to\infty} f(x) = b$

Example 2: Let $p(x) = -4x^5 + 2x^3 + 3x$

a) Find $\lim_{x\to\infty} p(x)$

$= \lim_{x\to\infty} (-4x^5 + 2x^3 + 3x)$

$= \lim_{x\to\infty} -4x^5 = \boxed{-\infty}$

really large pos #

*we know this limit will be $\pm\infty$.
we could plug in large values or
we could just look @ the leading term.

b) Find $\lim_{x\to-\infty} p(x) = \lim_{x\to-\infty} (-4x^5 + 2x^3 + 3x)$

$= \lim_{x\to-\infty} -4x^5 = \boxed{\infty}$

large negative #'s

Limits of Polynomial Functions at Infinity

If $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, $a_n \neq 0$, $n \geq 1$

then

$$\lim_{x\to\infty} p(x) = \lim_{x\to\infty} a_n x^n = \pm\infty$$

and

$$\lim_{x\to-\infty} p(x) = \lim_{x\to-\infty} a_n x^n = \pm\infty$$

Example 3: Determine the following limits:

a) $\lim_{x\to\infty} \frac{7x+12}{x^2+10x+5} = \lim_{x\to\infty} \frac{7x}{x^2} = \lim_{x\to\infty} \frac{7}{x} = \boxed{0}$

b) $\lim_{x\to-\infty} \frac{3x+4}{2x+1} = \lim_{x\to-\infty} \frac{3x}{2x} = \lim_{x\to-\infty} \frac{3}{2} = \boxed{\frac{3}{2}}$

c) $\lim_{x\to-\infty} \frac{x^2+7x+12}{x+5} = \lim_{x\to-\infty} \frac{x^2}{x} = \lim_{x\to-\infty} x = \boxed{-\infty}$

Finding limits at infinity for a rational function, $f(x)$:

Look for the highest degree of x:

1. If it is in the denominator, then $\lim_{x\to\pm\infty} f(x) = 0$
2. If it is in the numerator, then $\lim_{x\to\pm\infty} f(x) = \pm\infty$
3. If the degree of the polynomial in the numerator and denominator is the same then $\lim_{x\to\pm\infty} f(x) =$ the ratio of the leading coefficients

Example 4: Find all horizontal and vertical asymptotes of the functions below.

a) $f(x) = \frac{3x+2}{x-4}$ Domain: $x \neq 4$

Horizontal Asymptotes

$\lim_{x\to\pm\infty} \frac{3x+2}{x-4} = \lim_{x\to\pm\infty} \frac{3x}{x}$

$= \lim_{x\to\pm\infty} 3$

$= 3$

$\boxed{y=3}$

Vertical Asymptotes

$f(4) = \frac{3(4)+2}{4-4} = \frac{\text{nonzero \#}}{0}$

$\boxed{x=4}$

b) $g(x) = \dfrac{x^2+9}{x}$ Domain: $x \neq 0$

Vertical Asymptote

$g(0) = \dfrac{0^2+9}{0} = \dfrac{\text{nonzero \#}}{0}$

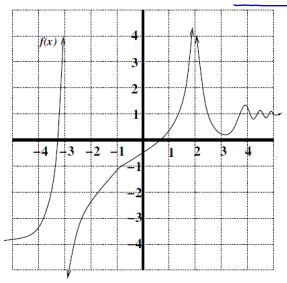
$\boxed{x=0}$

Horizontal Asymptotes

$\lim_{x\to\pm\infty} \dfrac{x^2+9}{x} = \pm\infty$

$\boxed{\text{No horizontal asymptotes}}$

Example 5: Find the limits at infinity, infinite limits, and asymptotes for the function f whose graph is shown.



$\lim_{x\to\infty} f(x) = 1$

$\lim_{x\to-\infty} f(x) = -4$

$\lim_{x\to-3^-} f(x) = \infty$

$\lim_{x\to-3^+} f(x) = -\infty$

$\lim_{x\to 2} f(x) = \infty$

Vertical Asymptotes

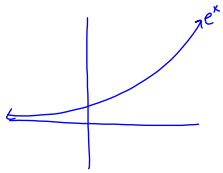
$x=-3,\ x=2$

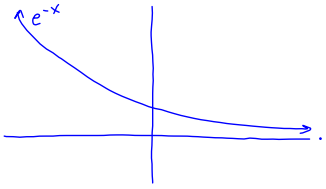
Horizontal Asymptotes

$\boxed{y=1}$ ↑ in the ∞ direction

$\boxed{y=-4}$ ↑ in the $-\infty$ direction

Example 6: Let's examine the behavior of e^x and e^{-x} as x approaches positive and negative infinity.





$\lim_{x\to-\infty} e^x = 0$

$\lim_{x\to\infty} e^x = \infty$

$\lim_{x\to-\infty} e^{-x} = \infty$

$\lim_{x\to\infty} e^{-x} = 0$

Example 7: Find all horizontal and vertical asymptotes of $f(x) = \dfrac{6e^x}{1-4e^x}$ Domain: $x \neq \ln(1/4)$

Horizontal Asymptotes

$\lim_{x\to\infty} \dfrac{6e^x}{1-4e^x} = \lim_{x\to\infty} \dfrac{6e^x}{-4e^x} = \lim_{x\to\infty} -\dfrac{3}{2} = -\dfrac{3}{2}$

$\boxed{y = -\frac{3}{2} \text{ in the } +\infty \text{ direction}}$

$\lim_{x\to-\infty} \dfrac{6e^x \to 0}{1-4e^x \to 0} = \dfrac{0}{1-0} = \dfrac{0}{1} = 0$

$\boxed{y=0 \text{ in the } -\infty \text{ direction}}$

Vertical Asymptotes

$1-4e^x = 0$

$1 = 4e^x$

$\ln(\frac{1}{4}) = \ln(e^x)$

$\boxed{\ln(1/4) = x}$

↑ vertical asymptotes

$f(\ln(1/4)) = \dfrac{6e^{\ln(1/4)}}{1-4e^{\ln(1/4)}} = \dfrac{\text{nonzero \#}}{0}$

Section 5.3 Highly Suggested Homework Problems: 1-17 (odd), 21, 27, 31